Engineering Innovation Challenge 2023

Flood-Proof Roads

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Jason Abi Chebli, Phan Tuan Kiet Le, William Wang

Monash University, Clayton, Australia

Abstract

Floods have become a challenge in many built-up areas due to increased urban development and uncertain climate conditions. Singapore, with its low-lying geography and heavy rainfall, experiences regular flash flooding. This issue correspondingly exists globally, posing significant risks in casualties and infrastructural damages. A solution of flood-proof roads was explored to replace traditional roads in flood-prone areas, particularly those in proximity to essential facilities such as hospitals, to prevent incidents such as that in Figure 1.

Figure 1: An ambulance is seen stuck in flood water near Changi fire station in Singapore on June 23, 2020 [1].

To achieve this objective, a comprehensive flood-proof road prototype, depicted in Figure 2, was designed and tested. The prototype comprises interconnected floating pontoons beneath porous road surfaces, equipped with a motorised deployment system. The system is ingeniously engineered to automatically detect rising water levels during flash floods and adapt the road's height, forming a functional floating surface above floodwaters.

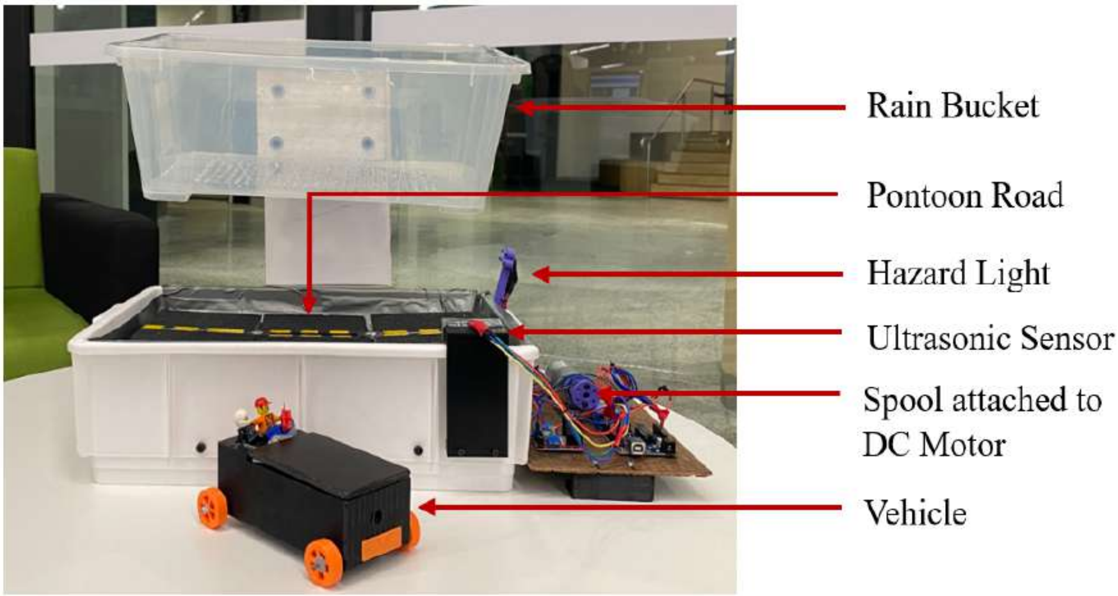

Figure 2: Final prototype design.

The prototype underwent rigorous evaluation in two main testing scenarios, as shown in Figure 3. Notably, the prototype demonstrated success in achieving critical engineering goals related to structural integrity, material selection, buoyancy, stability, adaptability to flood conditions, user experience, and safety.

Figure 3: Testing under dry and flash flooding conditions.

The positive outcomes of the prototype testing illustrate its potential as a resilient solution for safe commuting on flood-prone roads. As the effects of climate change continue to intensify, innovative solutions like flood-proof roads are vital in proactively preparing cities and regions for a future characterised by more frequent and severe flooding events.

Project Report

Research Problem

Climate change is impacting regions across the globe in different ways, and Singapore is no exception. The country has already witnessed a rise in annual rainfall and sea level, with rainfall increasing at 101 mm per decade from 1980 to 2016 and sea level steadily rising at a rate of 1.2 mm to 1.7 mm per year between 1975 and 2009 [2]. Experts predict that heavy rainfall intensity and frequency will continue to escalate, and sea levels could surge by up to 1 metre by the end of this century [2]. The rise in rainfall and sea levels means greater flooding risks, including river floods, flash floods, and coastal floods. At its current state, Singapore receives 100 to 300 mm of rain each month, with the average annual rainfall being 2,340 mm [3], and one-third of the country is less than 5 m above sea level [4]. Hence, to build resilience and mitigate the anticipated worsening of floods in the coming years, it is essential for Singapore to proactively future-proof its city from flooding.

In light of the escalating challenge posed by flash flooding in the present and future, a critical research question that needs to be addressed is: what innovative approach can be utilised to develop a resilient solution that enables safe commuting for drivers on flood-prone roads?

Project Objectives

The purpose of the prototype developed is to address the pressing issue of flash flooding, which significantly hinders safe commuting for drivers, particularly in emergencies and critical situations involving essential services. This project aims to devise a comprehensive prototype solution that enables drivers to traverse flood-prone roads, ideally replacing traditional asphalt roads that would typically be impassable due to flooding.

To achieve these objectives, several crucial engineering goals must be attained, including ensuring appropriate structural integrity, buoyancy, stability, material selection, adaptability to flood conditions, safety, and user experience. While anticipated outcomes of the prototype include a solution that is scalable and adaptable, allows for normal commutability during no-flood periods as well as enhanced commutability during floods. For a more comprehensive understanding of these objectives and outcomes, please refer to Appendix A: Detailed Project Outcome.

Final Proposed Prototype Design

To address flooding concerns, the proposed solution involved installing pontoons under porous roads capable of automatically detecting rising water levels and raising the road level during flash flooding. The small-scale prototype design comprised three main subparts:

- ➢ *Floatable pontoon roads:* The carefully engineered pontoons feature a 1:32 scale and rectangular shape, comprising 12 waterproof-coated FDM 3D printed PLA pontoon pads forming six blocks with interconnected loops for easy integration and road flexibility. Each pontoon can withstand the weight of a scaled-down 30,000 kg vehicle, ensuring engineering goals of structural integrity, material selection, and buoyancy and stability are achieved.
- ➢ *Pontoon deployment system:* To prevent floating away during flooding, each pontoon block has nylon strings attached to the under-compartment, fed through a pulley to the deployment system outside. An HC-SR04 ultrasonic sensor detects water levels, signalling an Arduino Uno, controlling an L298N Motor Driver to adjust the spool and strings, enabling the pontoons to float higher or lower. Yellow and red LED lights are also integrated to indicate different water levels and signal caution. Overall, the pontoon deployment system achieves the engineering goals of adaptability to flood conditions, safety, and user experience.
- ➢ *Simulating environment system:* The system includes a rain bucket above the road, simulating an even distribution of rainfall. A scaled vehicle with a modular weight ensures accurate testing.

Overall, the prototype demonstrates promising results, offering a potential flash flooding adaptation solution. To learn more about the subparts, the engineering calculations and the design decisions that led to the final design seen in Figure 4, please refer to Appendix B: Prototype Development.

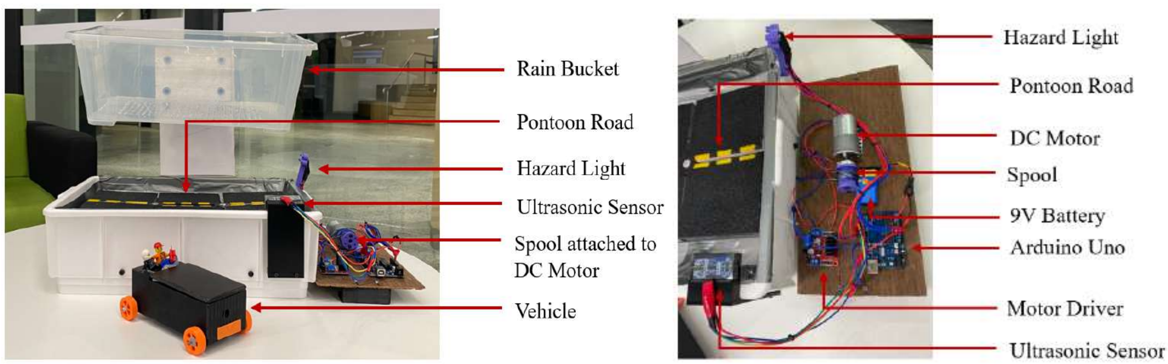

Figure 4: Final Prototype Design.

Experimentation

Testing Methodology

To validate the flood-proof road concept and assess its feasibility, two main testing scenarios were conducted on the small-scale prototype. For a detailed step-by-step guide to the testing methodologies, please refer to Appendix C: Prototype Experimentation.

Scenario 1: Commutability during Dry Conditions

The small-scale prototype requires testing under dry conditions to ensure functionality in all weather. This involves validating the structural integrity of the pontoon roads and the selected materials. If the road can support the weight of the small-scale vehicle without yielding, it confirms the appropriate material selection and achieves the first engineering goal of normal commutability. To test commutability during dry conditions, verify the vehicle's correct weight, and then place and move it along the pontoon road, noting traction and any yielding.

Scenario 2: Commutability during Flash Flooding Rainfall

To assess the material selection and the pontoon's design ability to handle flash flooding rainfall, the prototype must undergo wet condition testing. The objective is to verify that the system accurately detects rising water levels, raises the roads, and activates hazard lights while ensuring water flow beneath the roads. To test commutability during flash flooding rainfall, pour the specified amount of water into the rain bucket and verify that the yellow lights begin to flash and the motor releases enough string for the roads to float. Add more water to verify if the red hazard lights are triggered while the pontoons maintain buoyancy. Finally, place and move the vehicle along the pontoon road to assess the pontoons' capacity to withstand the weight and keep the vehicle above water level, while noting any observations.

Results and Discussion

Under Scenario 1 testing, commutability during dry conditions, the vehicle's weight was appropriately 915.5 grams, as required, and the pontoon roads easily withstood this load. The pontoons made no observable noises or visible signs of yielding, and the anti-slip tape used to simulate asphalt grip provided sufficient traction for the vehicle. Furthermore, as expected, no hazard lights were signalling since there was no water on the roads, and they were not floating. Overall, this validates the engineering goals of structural integrity and appropriate material selection.

In Scenario 2 testing, commutability during flooding rainfall, as the specified water amount was added, the system accurately detected the water level increase. The motor released enough string to enable the pontoons to float, and the yellow hazard lights began flashing. Most of the water flowed through the small gaps, effectively preventing flash flooding from occurring on top of the roads. When additional water was added, the red hazard lights signalled extremely high water levels, and sufficient string was released to maintain buoyancy. However, as the water dissipated, the system returned to a less severe state, resulting in yellow flashing and the motor retracting some string, in line with the design.

During driving over the pontoons, despite some slight water permeating the top (which is to be expected), sufficient traction was maintained, and the roads remained buoyant. The vehicle never got submerged in the floodwater, validating the engineering goals of buoyancy, stability, adaptability to flood conditions, safety, and user experience.

Overall, the prototype design of the pontoon road system demonstrated exceptional effectiveness in both dry and flash flooding conditions, successfully fulfilling the engineering objectives of normal commutability during no-flood periods and enhanced commutability during floods. Moreover, the final engineering goal of scalability and adaptability was also clearly attained. However, further testing and refinement are necessary due to some limitations in the design and testing process, including:

1. *Testing different pontoons:* Evaluating various pontoon patterns and volumes is crucial as they may yield different dynamic outcomes, some of which may be more effective or less desirable. Therefore, conducting tests with different configurations is essential to explore and optimise the prototype's performance.
2. *Yielding and life cycle testing:* The ultimate yielding of the pontoons and their life cycle were not tested, and these aspects should be considered in future tests for a comprehensive evaluation.
3. *Uncertain water levels:* Confirming water levels solely based on the ultrasonic sensor may have limitations. Implementing an alternative validation method, such as a small-scale flood gauge, would enhance data accuracy.

4. *Assessing different rainfall conditions:* More comprehensive testing is required to evaluate the performance under various rainfall water amounts and flow rates, utilising the rain bucket setup.
5. *Assessing faster-moving cars:* Conducting further testing to determine achievable vehicle speeds and their impact on the roads is essential to simulate real-world traffic conditions.
6. *Considering scalable materials:* While the current materials serve as proof of concept, further investigation with larger budgets is necessary to identify practical and scalable materials for the pontoons.

Conclusion

The success of the small-scale prototype of the floatable road concept represents a significant step forward in addressing the pressing issue of flash flooding in Singapore. The experiment successfully validated the engineering goals of structural integrity, appropriate material selection, buoyancy and stability, and adaptability to flood conditions. During dry conditions, the pontoon roads exhibited high structural integrity, withstanding the weight of the vehicle without yielding. In scenarios of flash flooding rainfall, the system accurately detected rising water levels and activated hazard lights, effectively preventing flash flooding on the roads by raising the level of the roads and ensuring safe commuting for drivers.

The prototype's positive results underscore its potential as a resilient solution to enable safe commuting on flood-prone roads. Its adaptability and scalability offer promising prospects for broader implementation in Singapore and other flood-prone regions. However, further testing and refinement are necessary to overcome the limitations observed, including the testing of different pontoons, yielding and life cycle testing of the pontoons, assessing faster moving cars and different rainfall conditions, and the need to consider scalable materials. Based on these findings, future experiments should explore these limitations.

Overall, as the effects of climate change continue to worsen and flooding becomes more prominent and intense, in Singapore and across the world, this issue must be addressed proactively. Developing innovative and resilient solutions, like the pontoon roads, can play a crucial role in building climate resilience and ensuring the safety and well-being of current and future communities in the face of increasing environmental challenges.

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Appendix A: Detailed Project Outcome

Current Flooding Solutions Available

Current solutions that Singapore has implemented include green roofs, retention ponds, detention tanks, porous pavements, rain gardens, diversion canals, canal improvements, minimum platform and crest levels, reservoirs, and flood barriers [5]. Although these solutions have helped, they have not effectively eliminated flooding, with floods still occurring in Singapore. Furthermore, these floods are dangerous and can significantly impact the operations of a city and the lives of its people. For example, road closures resulting from flooding can prevent ambulances from delivering patients to hospitals promptly, as well as individuals may take the risk of driving through flood waters, even though it is unsafe to do so. Moreover, flooding is a major concern not only in Singapore but also in many parts of the world and as a result, a solution would not only benefit Singapore but also numerous cities worldwide.

Research Question

What innovative approach can be utilised to develop a resilient solution that enables safe commuting for drivers on flood-prone roads?

Aim

The overall aim of the project is to create a solution that will allow drivers to safely commute over roads that would usually be closed due to flooding. It is hoped that the solution could be implemented instead of traditional asphalt roads, especially on roads that are usually prone to flooding as well as critical roads such as roads near hospitals.

Hypothesis

It is hypothesised that incorporating pontoons underneath porous roads would allow for water to seep through while enabling the roads to float above, which ultimately facilitates the safe commuting of drivers on roads that are commonly closed during flooding incidents.

Engineering Goals

The engineering goals that need to be achieved to produce a viable solution include:

1. *Structural Integrity:* Ensure that the pontoon road design can withstand the anticipated loads and stresses, including the weight of vehicles, water pressure, and dynamic forces during floods.
2. *Buoyancy and Stability:* Design the pontoons to provide sufficient buoyancy to keep the road afloat during flood events, maintaining stability and preventing any tilting or instability.
3. *Material Selection:* Identify appropriate materials for the pontoon structure and porous road surface that can withstand water exposure, provide sufficient strength, and ensure durability over the expected lifespan of the road.
4. *Adaptability to Flood Conditions:* Design the pontoon road system to be responsive and adaptable to varying flood conditions, including water levels, flow rates, and duration, to maintain functionality and safety.
5. *Safety and User Experience:* Prioritise the safety of drivers, pedestrians, and other road users by incorporating features that enhance visibility, provide adequate road grip, and mitigate potential hazards associated with the pontoon road design.

Expected Outcomes

On completion of the project, expected outcomes include:

1. *Scalable and Adaptable:* The goal is to build an accurate small-scale working prototype of a section of a road pontoon. It is expected that this pontoon road prototype is adaptable and scalable to fit various flood-prone areas and road networks. The expected outcome is a flexible solution that can be customised based on the specific requirements and conditions of different locations.
2. *Normal Commutability during No-Flood Periods:* As this solution is intended to be in place at all times, even when flooding does not occur, the prototype is expected to demonstrate that it is functional under normal, non-flood conditions.
3. *Enhanced Commutability during Floods:* The prototype is expected to demonstrate its ability to prevent flooding on roads by allowing water to seep below the road surface and safely raising the roads when flooding occurs. This is expected to allow

small-scaled cars to traverse the roads safely. Furthermore, it is anticipated that as water levels lower, the roads will gradually lower as well.

Sustainable Development Goal

The United Nations set out 17 Sustainable Development Goals to be completed by 2030 [6] as shown in Figure 5.

Figure 5: UN’s 17 Sustainable Development Goals. [6]

Addressing flash flooding is essential as a lack of flood resilience can impede or reverse progress towards achieving the Sustainable Development Goals [7]. Additionally, creating flood-proof roads can help achieve the 9th sustainable development goal which focuses on building resilient infrastructure. More specifically, this solution can address Target 9.1, which focuses on developing quality, reliable, sustainable and resilient infrastructure [6], overall improving the quality of life for people around the world.

Appendix B: Prototype Development

Scaling

A real-scale prototype of flood-proof roads would be near impossible with the budget and time constraints. As such, a small-scale prototype of the pontoon road is necessary to illustrate a proof of concept. To determine the small-scale dimensions for the pontoon, we require some equations and need to perform some calculations as if it were a real scale prototype.

Pontoons are buoyant structures typically filled with air [8]. Achieving the engineering goal of buoyancy and stability required careful consideration of the pontoon's design and air volume. To determine the optimal scale for the prototype, we will assume the volume of air inside the pontoon to be that of a rectangular prism, which can be calculated using Equation 1.

$$V = lwh \tag{1}$$

Where V is the fluid volume, l is the length, w is the width, and h is the height of that rectangular prism.

Regarding buoyancy, Archimedes' principle states that the buoyant force can be calculated using Equation 2 [9].

$$F_b = -\rho g V \tag{2}$$

Where F_b is the buoyant force, ρ is the fluid density, g is the gravitational acceleration, and V is the fluid volume.

Substituting Equation 1 into Equation 2, we get Equation 3.

$$F_b = -\rho g l w h \tag{3}$$

The weight force of the vehicle is outlined in Equation 4 [9].

$$F_W = mg \tag{4}$$

Where m is the mass of the vehicle and g is the gravitational acceleration.

Now, for the vehicle and the pontoon road to float, the sum of weight forces should balance out with the buoyant force, or in other words the sum of the forces in the y-direction should be zero, as outlined in Equation 5. When substituting it in we get an expression that relates the height to the mass and the other dimensions of the pontoon as outlined in Equation 6.

$$\sum F_y = 0 \tag{5}$$

$$\begin{aligned} mg - \rho glwh &= 0 \\ mg &= \rho glwh \\ m &= \rho lwh \\ h &= \frac{m}{\rho lw} \end{aligned} \tag{6}$$

Typical lane width in Singapore is often 3.2m or more [10], hence it is reasonable to assume that the lane width is 3.2m. Now, let's assume that for the prototype, the desired width of one lane is 0.1m and the length is 0.12m. From this we could work out the scaling factor as outlined in Equation 7.

$$\begin{aligned} Scale &= \frac{w_{real}}{w_{scaled}} \\ Scale &= \frac{3.2}{0.1} \\ Scale &= 32 \end{aligned} \tag{7}$$

Now that the scale of 32 is known, the length of the real scale pontoon can be calculated as outlined in Equation 8.

$$\begin{aligned} l_{real} &= Scale \times l_{scaled} \\ l_{real} &= 32 \times 0.12 \\ l_{real} &= 3.84\, m \end{aligned} \tag{8}$$

We know that in Singapore, the heaviest vehicle type is a Very Heavy Goods Vehicles, which has a weight exceeding 16,000kg [11]. For testing purposes, a weight of 30,000kg will be assumed as the maximum weight. We assume the density of rainfall to be the density of water, which is approximately $1000\ kg/m^3$ [9]. As such, using Equation 6, we can determine the real height of the pontoon required.

$$\begin{aligned} h_{real} &= \frac{m_{real}}{\rho\ l_{real}\ w_{real}} \\ h_{real} &= \frac{30,000}{(1,000)(3.84)(3.2)} \\ h_{real} &\approx 2.44\ m \end{aligned}$$

This means that for a real scale pontoon to hold a 30,000kg vehicle, it would need to have a volume of 3.2m by 3.84m by 2.44m. This assumes that compared to the vehicle, the pontoon's weight is so small, it can be considered negligible. Knowing the real height, we can then calculate the scaled height using Equation 9.

$$h_{scale} = \frac{h_{real}}{Scale} \qquad (9)$$
$$h_{scale} = \frac{2.44...}{32}$$
$$h_{scale} \approx 0.076m$$

Furthermore, we can calculate the mass of the small scale vehicle, from a modification of Equation 6.

$$m = \rho lwh$$
$$m = (1000)(0.12)(0.1)(0.076...)$$
$$m = 0.9155\ kg$$

This means that the scaled-down prototype needs to be: 0.1m by 0.12m by 0.076m and withhold a vehicle load of 0.9155kg to be scalable. These numbers were run through MATLAB and a range of different values were imputed to determine the most reasonable values. The ones presented here were the ones we finalised with.

Note that the scaling factor for the dimensions is 32, while for the mass it is $32^3 = 32768$. This means the scaling is not consistent, however, we believe the discrepancy in scaling is acceptable for this initial proof-of-concept prototype for several reasons. First, real-scale testing would involve a significantly larger and more complex setup, requiring different materials and resources. It would be impractical and costly to replicate the exact scaling for this preliminary stage. Second, the purpose of this small-scale prototype is to demonstrate the feasibility and effectiveness of the flood-proof road concept in a controlled environment. By conducting tests on a smaller scale, we can validate the fundamental principles and engineering goals without incurring the expenses and logistical challenges associated with a full-scale implementation. Additionally, the materials selected for this initial prototype are appropriate for showcasing the concept's potential and functionality. While they may not be suitable for large-scale implementation, they serve the purpose of validating the core principles of the design, including structural integrity, buoyancy, and adaptability to flood conditions.

Pontoon Road

The pontoon length was chosen to be lane width to simplify dynamic forces by allowing only for at most 2 cars to be on one pontoon at a time.

The final pontoon design, seen in Figure 6, consisted of 12 individual pontoon pads, which would connect to make 6 individual air-filled pontoon blocks, all linked by two bolts. Small gaps between pontoon blocks were included in the design to allow for water to flow underneath it - preventing flash flooding - as well as in theory allow for the blocks to expand and shrink slightly due to temperature changes throughout the year. It also includes the interconnect loops, allowing the ease of integration with other units while providing structural flexibility to stay afloat in water with various flow rates and permeability under rainy conditions.

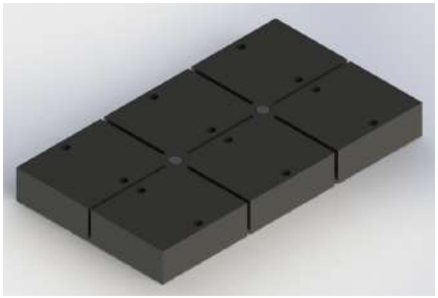

Figure 6: Final Pontoon Design.

For prototyping purposes, each pontoon pad is made from 3D printed PLA, which would be connected via waterproof silicone sealant and reinforced by tape. Additionally, as 3D printed parts are not waterproof, a waterproofing coat is also added on top. Other methods were considered when manufacturing these, such as resin printing (which is more watertight), but FDM printing was finalised as it was the quickest, easily available and most affordable as well as had a high structural integrity to withstand loads. As each pontoon pad took at least 6 hours to print, to validate our concept, an initial test was done on one pontoon block, to make sure it floats and holds some forces, as shown in Figure 7.

Figure 7: Initial Pontoon Design Floating.

With our initial design floating perfectly and validated, we proceeded to print the rest of and perform the same procedure, including silicone sealant, taping and waterproof spraying. We then completed the pontoon design by adding a water-resistant layer of anti-slip grip tape on top to simulate the grip from asphalt on normal roads and painted centre road lines. This process can be seen in Figure 8.

Figure 8: Development of the Pontoon.

Pontoon Deployment System

To ensure that the pontoons do not simply drift away during flash flooding, a pontoon deployment system was essential. Each pontoon block has nylon braided string (that was hot-glued on) to ensure that the pontoons do not drift away. However, more than that, the deployment system consists of a spool that was attached to a DC motor that would release an adequate amount of string if water levels increased, as well as retract an adequate amount of string, as the water level dissolves underneath. This DC motor that determined the necessary string length was controlled by the Arduino Uno using HC-SR04 ultrasonic sensor and L298N Motor Driver. Initially, the electricals were going to be right underneath the pontoons, however, due to safety concerns of water leaking, the electricals were placed outside, and the nylon string was directed via a pulley installed underneath as can be seen in a segment of Figure 10. Overall, this pontoon deployment system achieved the engineering goal of making the solution adaptable to flood conditions. Additionally, to ensure the safety and user experience of drivers (another engineering goal), a hazard light system was designed and incorporated that would indicate when the roads were floating and driving had to be more careful. The idea is that when the pontoons float at a certain level, a yellow light will appear to indicate that it is floating and to be careful. If the water level continued to increase to a high threshold, red lights would indicate, indicating that it is quite dangerous and that vehicles should drive extremely slowly and carefully.

We initially set up our electronics on a breadboard, which was later transformed into a proper platform. This involved performing proper cabling, crimping, soldering, heat shrinking, and powering to ensure its reliability and independence. The development progress and transformation can be seen in Figure 9.

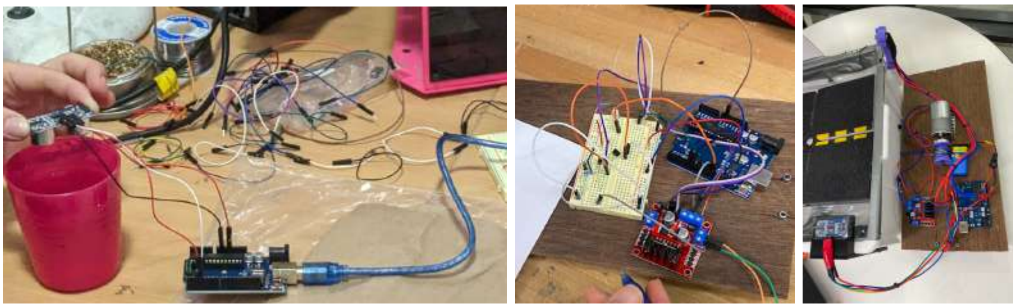

Figure 9: Development of the Pontoon Deployment System.

Simulating Environment System

The whole system was contained in a plastic container. A plastic plate was placed inside the container on brackets, to simulate the ground. And a pulley was installed underneath to feed the string to the motor outside. Additionally, another plastic container was used to replicate intense rain with holes drilled out throughout the container to simulate the rate and distribute the water evenly across the whole road. Furthermore, to validate the structural integrity and accuracy of our prototype, a 3D-printed scaled-down vehicle made with appropriate scaled weight was used to simulate the maximum weight of a vehicle driving over the roads. The design of the vehicle allowed for weight to be added internally and so sand was added internally until the total weight was 0.9155kg. Overall, the different aspects of the simulated environment can be seen in Figure 10.

Figure 10: Different Aspects of the Simulating Environment.

Material List

To develop the prototype, the materials required can be found in Table 1.

Table 1: Materials List Summary.

Quantity	**Material**	**Cost**	**Source**
1	Roll of PLA	$27.96	[12]
1	Silicon Sealant	$16.80	[13]
1	Duct-Tape	$3.30	[14]
1	Anti-Slip Tape	$10.71	[15]
1	Waterproof Spray Paint	$14.12	[16]
1	Yellow Liquid Paint Marker Pen	$4.98	[17]
1	Zip lock bags	$1.60	[18]
1	Acrylic sheet	$59.31	[19]
1	Wooden Block	$21.75	[20]
1	Nylon String	$5.50	[21]
1	Pulley	$3.75	[22]
1	Angle brackets 20 Pack	$6.25	[23]
1	Sand Bag	$10.40	[24]
1	Arduino Uno	$34.95	[25]
1	Ultrasonic Distance Sensor Breakout For Arduino	$7.95	[26]

1	L298N Dual Motor Module For Arduino	$16.95	[27]
1	DC motor	$29.95	[28]
1	Power Cable/Wires	$8.20	[29]
2	Red LEDs	$0.70	[30]
2	Yellow LEDs	$0.70	[31]
1	Assorted Nuts and Bolts	$20	[32]
	Total Cost:	$305.83	

Using these materials, you will require general hand tools, including hand drills, hacksaws, Allen keys, socket sets, caulk gun, hot-glue gun, soldering irons, heat shrink guns, and crimping tools. You will also require access to a FDM 3D Printer as well as the licensing for software programs such as SolidWorks, Cura, MATLAB, and Arduino.

Reproducible Manufacturing Steps for the Prototype

Pontoon Road

1. Create the CAD model on Solidworks based on the scaled dimensions and pre-defined design constraints.
2. Export the files as an .STL format and slice them in Cura to generate the g.code for 3D printing.
3. Proceed to 3D print the 12 pontoon pads. Please note that this process may take several days of continuous printing.
4. Once all the necessary parts are printed, begin the assembly process to verify which parts fit together. Use silicone sealant to caulk the pontoon halves together and allow them to set for a day.
5. Next, apply a waterproof base coat, followed by a top coat after ensuring sufficient setting time. Always remember to wear a mask and work in a well-ventilated area, as fumes may be toxic. Take care to allow adequate time for the coatings to set before proceeding further.
6. The next day, add a final layer of tape over the joints of all the pontoon blocks for additional reinforcement.

7. Subsequently, apply anti-slip grip tape on top of the pontoon surface to provide traction for cars to drive on. Additionally, use a yellow liquid paint marker pen to add road lines for a more polished presentation effect.
8. Finally, measure out enough string for deployment and use a hot glue gun to adhere it to the middle of the under-compartment of each pontoon block.

Pontoon Deployment System

1. Prototype the control system using a breadboard, Arduino Uno, and an Ultrasonic sensor to test for control signals.
2. Integrate the system with a motor driver, DC motor, and LEDs to test for the overall control of the software program.
3. Design the LEDs hazard lights mount, the ultrasonic sensor mount and the spool that will connect with the motor.
4. 3D print the LEDs hazard lights mount and the spool.
5. Finalise the electrical components and wires by soldering and crimping connectors between input/output components and the Arduino controller.
6. Create a mounting plate for all of the components located outside of the container.

Simulating Environment System

1. Design a CAD model on SolidWorks for the vehicle, ensuring it is to scale and hollow to allow for modular weight to be added inside. 3D print this CAD model.
2. Measure the dimensions of the pontoon road and acquire a container that is large enough (but not excessively large) for the pontoon road to fit in. Ensure the container has sufficient depth.
3. Drill holes within the container to mount angle brackets, which will hold the ground.
4. Mount the pulley in the centre base of the container.
5. Cut an acrylic board to fit within the container and rest on top of the angle brackets, acting as the base ground on which the pontoons will sit. Drill a hole through the centre of the board to allow the strings of the pontoons to feed through.
6. Cut plastic zip lock bags and tape them down along the edges of the acrylic board to minimise water leakage through the sides.

7. Measure and drill evenly spaced holes through another plastic container, which will serve as the rain bucket.
8. Cut a combination of acrylic and wood to mount the rain bucket to the side. Drill holes and fasten bolts to secure it in place. When determining the right distance for the rain bucket, place the pontoons inside and ensure that the drilled holes cover the pontoon road.

Final Assembly Step

1. Feed the string through the hole in the ground and around the pulley, then guide it out through a hole on the side of the container. Slowly lower the road to ensure that the string remains continuously in tension.
2. Attach the string securely to the 3D printed spool using tape, ensuring a firm connection.
3. Connect the spool to the DC motor, ensuring it is firmly attached to enable proper rotation and winding of the string.

Appendix C: Prototype Experimentation

The following methodology is a step-by-step guide to test the two different scenarios for the prototype.

Scenario 1: Commutability during Dry Conditions

Setup

1. Use a high-precision laboratory scale to measure the weight of the PLA vehicle.
2. Note down the current weight and determine the remaining weight needed to meet the required weight.
3. Pour sand into a zip lock container and zip it up.
4. Weigh the sand, and if it is overweight, remove some sand; if it is underweight, add more sand.
5. Re-weigh the sand after making adjustments.
6. Repeat the process until the weight matches the required weight.
7. Once the weight is correct, place the zip lock bag with sand inside the interior part of the vehicle by lifting up the lid.
8. Confirm the weight by re-weighing the entire setup together.
9. Have someone record the following test for documentary purposes.

Testing

1. Position the vehicle on the pontoon road carefully.
2. Note down any initial observations.
3. Using your hand, drive the vehicle across the road, and even perform 3-point turns.
4. Observe the structural integrity of the pontoon road while moving the vehicle.
5. Note any signs of yielding or structural instability, and record any observations or indications of road deformation.
6. Assess the traction of the pontoon road material.
7. Observe how the vehicle behaves in terms of surface traction and grip when stopping and starting on the road.

Scenario 2: Commutability during Flash Flooding Rainfall

Setup

1. Ensure that the flood-proof road prototype, the scaled car, and a bucket or bowl filled with water (with a specified amount measured using a measuring cup) are available and properly arranged as shown in Figure 4.
2. Verify that the holes from the rain bucket align and cover the full length of the road when observed from the top.
3. Have someone ready to record the testing process on video or set up cameras to capture the experiment as it may occur quickly.

Testing

1. Pour the specified amount of water into the rain bucket while someone holds it.
2. Observe the system's response to the added water, which should include the motor rotating, the roads beginning to float, and the yellow LED hazard lights flashing on and off. Record whether these observations occur as expected.
3. Pour additional specified water into the rain bucket to simulate a more intense flash flooding scenario.
4. Observe whether the red LED hazard lights begin to flash, the motor continues to rotate, and the road maintains its floating position.
5. As the water starts to dissipate downward, the water level should lower, causing the motor to retract some string, the pontoons to lower, and the lights to revert to yellow.
6. Note down whether these observations occur during the decrease in flash flooding.
7. Using your hands, test how easily the pontoons sway away from their set position and record these observations.

Vehicle Testing

1. Place the vehicle on top of the pontoon roads to observe if the weight causes the pontoons to sink.
2. Move the vehicle along the road and note whether there is sufficient surface traction.
3. Observe if the vehicle becomes submerged at any point during the test.

Appendix D: Risk Management Plan

The Risk Matrix that was used can be seen in Table 2.

Table 2: Risk Matrix [33].

Likelihood	**Consequences**				
	Insignificant *Risk is easily mitigated by normal day to day process*	**Minor** *Delay up to 10% of Schedule. Additional cost to 10% of Budget*	**Moderate** *Delays up to 30% of Schedule Additional cost up to 30% of Budget*	**Major** *Delays up to 50% of Schedule Additional cost up to 50% of Budget*	**Catastrophic** *Project Abandoned*
Certain >90% chance	High	High	Extreme	Extreme	Extreme
Likely 50%-90% chance	Moderate	High	High	Extreme	Extreme
Moderate 10%-50% chance	Low	Moderate	High	Extreme	Extreme
Unlikely 3%-10% chance	Low	Low	Moderate	High	Extreme
Rare <3% chance	Low	Low	Moderate	High	High

A risk management plan that was considered when undertaking this project can be seen in Table 3.

Table 3: Risk Management Plan.

Risk	**Likelihood**	**Consequences**	**Risk Level**	**Mitigation**
Fires and Burns from 3D printing and	Moderate	Minor	Moderate	Adequate ventilation, have a fire extinguisher on standby, use PPE

soldering				
Electrical Shock from circuitry	Unlikely	Moderate	Moderate	Disconnect power before touching, use PPE, insulate any exposed wires
Toxic Fumes from Soldering	Moderate	Minor	Moderate	Adequate ventilation, use PPE, avoid unnecessary idle time and take breaks
Water Damage	Moderate	Moderate	High	Ensure waterproof enclosures, proper cable management, proper insulation and sealing